An Iterative Change Detection Method Based on Robust PCA for Small SAR Image Datasets

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***Abstract*—This article introduces an iterative change detection (CD) method for synthetic aperture radar (SAR) imagery based on robust low-rank and sparse matrix decomposition, specifically robust principal component analysis (RPCA). Instead of using fixed pairs of reference and surveillance images, as are typically used in conventional RPCA-based CD methods, our method progressively integrates multiple reference images and adaptively updates the regularization parameter at each iteration. This iterative scheme improves the separation between static clutter and sparse changes, significantly reducing the false alarm rate (FAR). The proposed iterative CD method is also explored using an extension of RPCA to tensors, known as tensor robust principal component analysis (TRPCA), thereby enabling its application to multitemporal and multidimensional SAR datasets. The experiments were conducted using real SAR data from the Swedish CARABAS-II system. The results showed that our iterative scheme outperforms noniterative CD methods based on a pair of SAR images and achieves competitive results when compared to CD methods using SAR image stacks, indicating that the iterative approach can be an effective solution in scenarios where SAR image availability is limited and SAR acquisitions are expensive.**

***Index Terms*—CARABAS-II, iterative change detection (CD), robust principal component analysis (RPCA), synthetic aperture radar (SAR), tensor RPCA (TRPCA).**

NOMENCLATURE

IA	Imaged area.
ADMM	Alternating direction method of multiplier.
AFRL	U.S. Air Force Research Laboratory.
ALM	Augmented Lagrange method.
ATR	Automatic target recognition.
CARABAS	Coherent all radio band system.
CD	Change detection.
CD-RPCA	Change detection based on RPCA.
CD-TRPCA	Change detection based on TRPCA.
DFT	Discret Fourier transform.
DT	Detected targets.
FA	False alarm.
FAR	False alarm rate.
FFT	Fast Fourier transform.
FOA	Swedish National Defence Research Establishment.
FOPEN	Foliage penetration.
GMTI	Ground moving-target indication.
GPR	Ground penetrating radar.
KT	Known targets.
PC	Principal component.
PCA	Principal component analysis.
PCP	Principal component pursuit.
PD	Probability of detection.
RPCA	Robust principal component analysis.
SAR	Synthetic aperture radar.
SDMS	Sensor data management system.
SVD	Singular value decomposition.
TNN	Tensor nuclear norm.
TRPCA	Tensor robust principal component analysis.
T-SVD	Tensor singular value decomposition.
UWB	Ultra-wideband.
VHF	Very-high frequency.
WR	Wavelength-resolution.

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I. INTRODUCTION

IN RECENT years, space agencies have deployed a significant number of Earth-monitoring satellites, mainly equipped with SAR sensors [1], [2]. Furthermore, it's possible to observe interest among agencies in exploring SAR sensors coupled to autonomous aircraft systems, also known as drones [3], [4]. SAR systems stand out in remote sensing applications because they provide high-resolution images 24 hours a day, regardless of atmospheric conditions. The microwave signals transmitted by SAR sensors can penetrate clouds, fog, and, in some cases, leaves, snow, and sand [5]. These characteristics

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are extremely important for applications such as urban and forest monitoring, vegetation mapping, ocean topography, and object detection and recognition [5], [6].

CD has proven to be one of the most efficient applications using SAR sensors [7], [8], [9], [10], [11], [12]. CD methods seek to identify spatial changes caused by natural or anthropogenic phenomena using multitemporal images. Changes may represent things that emerged or disappeared in the scene over time, enabling understanding of interactions between humans and nature, which in turn helps decision-making in many scenarios, such as urban development, environmental monitoring, defense, and security, among others [13].

There are currently different approaches to performing CD, including object- or pixel-based methods, supervised or unsupervised, and machine learning techniques [14]. Some of them are based on image pairs [15], [16], while more recent studies explore the use of SAR image stacks [17], [18]. One can also observe a high interest in iterative CD approaches [19], [20], [21], [22], [23], [24], [25]. Such processes usually outperform noniterative ones because the CD method's parameters are updated across iterations. In addition, we can observe different methods applied to synthetic SAR images. This approach is mainly observed in ATR SAR applications [26], [27].

Recent artificial-intelligence-based methods have also been increasingly explored for remote sensing and SAR CD. These approaches include convolutional neural networks, Siamese networks, autoencoder-based models, dual-domain networks, and other deep-learning architectures [28], [29], [30]. Such methods can learn complex spatial, semantic, and multiscale representations from multitemporal images, leading to strong performance in several CD scenarios. However, their performance is usually dependent on the availability of representative training data, reliable labels, and sufficient computational resources for model training and validation [29]. These requirements can be restrictive in practical SAR scenarios, especially when only a small number of well-coregistered acquisitions are available or when labeled change/no-change samples are scarce.

Recently, Schwartz et al. [31] showed that the CD problem can be seen as a signal separation problem of a sparse nature. The authors proposed a CD method based on RPCA via PCP to detect targets in a surveillance image using SAR image pairs of the same scene acquired at different times. RPCA decomposes the input matrix into low-rank and sparse components; thus, the motivation for the study lies in the sparse nature of the targets, which the sparse content of the decomposition must represent. This method can be considered a baseline for developing CD methods based on RPCA. Ramos et al. [32] extended [31] by exploring RPCA via PCP on small stacks of SAR images to detect targets in surveillance images. In this scenario, two forms of diversity were jointly explored: temporal and flight-heading diversities. Temporal diversity is explored by increasing the number of SAR images acquired with the same flight heading in the stack, whereas diversity in flight headings is explored through SAR image stacks formed from images from different flight headings.

Traditional CD methods, either for optical images or SAR images, are based on matrix approaches [33]. In fact, matrix-based strategies do not exploit the multidimensionality of the data since vectorization is generally necessary. An interesting strategy is to propose methods based on tensors, i.e., a generalization from scalars, vectors, and matrices into multidimensional spaces. The main motivation for this approach is that big new data from modern heterogeneous sensor modalities have a multidimensional character and are naturally represented by tensors [34]. Many recent CD methods have adopted tensor-based approaches, mainly using hyperspectral images [35], [36], [37]; however, this is still a relatively new topic for SAR images. Recently, Ramos et al. [38] have proposed to perform CD based on the TRPCA via a new TNN [39], which aims to exactly recover a low-rank tensor from the tensor corrupted by sparse noise. The method was evaluated on SAR image pairs and stacks, and better detection results were obtained compared to matrix-based approaches, since TRPCA allows exploration of the multidimensional characteristics of the stack, i.e., temporal and spatial diversity.

The studies mentioned above show that stack-based CD approaches that explore more than two coregistered images acquired over the same area of interest perform better. In the context of RPCA and TRPCA, obtaining more SAR images is even more important. First, these techniques are sensitive to input content and are generally used in image/video processing applications involving large numbers of small optical images. Since these applications consider many images as input, e.g., 100, 1000, or 5000 images, then robust tools accurately separate sparse and low-rank content [40], [41]. Note that this scenario is totally different from CD in SAR images, where there are few samples of a large imaged region. Second, the SAR image is formed by the returns from different reflectors in the scene, such as vehicles, buildings, transmission lines, and fences. Any content that appears or disappears between two images will be considered sparse and, consequently, detected. This content might be related to an object of interest, such as vehicles that showed up in the scene. Otherwise, it might be related to objects sensitive to the SAR platform's flight heading, poorly correlated images, or the speckle noise typically present in SAR images. Thus, the importance of having stacks of SAR images to differentiate these sparse detections stands out.

In recent years, the release of SAR systems has increased. In particular, space-based systems, such as Capella Space and ICEYE, are of great interest. Together with aircraft-based systems, these systems will add new capabilities, such as the new space-based systems with very high coverage and updates almost every hour. However, obtaining well-coregistered images of the same scene, i.e., with high image coherence, remains difficult [42]. For example, small changes in the speed and trajectory of the SAR platform can cause a static object to appear at different positions between images, leading to misalignment. In addition, purchasing many SAR images can be very costly, which motivates the use of iterative strategies that exploit the information contained in a few images.

Inspired by results obtained from iterative CD methods [21], [25], [43] and the use of SAR image stacks [22], [23], we are proposing an iterative CD method based on RPCA

decomposition. The proposed iterative approach is based on SAR image pairs and explores the low-rank and sparse matrices to create a new reference SAR image. The reference image obtained at each iteration and the original SAR images are used to apply RPCA, and the iterative CD method converges until a stopping criterion, λ_{max}, is reached.

From this approach, the targets in the surveillance image (i.e., the image you want to detect targets in) become sparse, thereby improving detection. In addition, the reference SAR image stack will eliminate possible changes related to the reference content, thus reducing the number of false alarms in the surveillance image. Recently, Yoon et al. [43] proposed RPCA on GPR in an iterative way. The method seeks to detect landmines and suppress ground reflections coming from the clutter, such as coins and stones. Initially, RPCA is applied to decompose the input radar signal into low-rank and sparse components. RPCA is reapplied, now to the sparse component, separating detections that remain stable over time, such as the signal of interest (i.e., landmines), from possible sparse false alarms from the ground. The results show that this strategy can preserve the target signal while eliminating undesired detections, thereby improving performance over previous methods.

Similar to RPCA [31], [32], and TRPCA [38] reference methods and others CD methods [44], [45], our CD method is evaluated on SAR images from the CARABAS-II system. The experimental results show that the proposed method outperforms recently proposed noniterative CD methods based on RPCA and TRPCA for SAR image pairs. In addition, the iterative scheme achieved competitive results compared with these methods when considering SAR image stacks. The major contributions of this article are listed as follows.

1) An iterative unsupervised CD method on SAR image pairs based on robust PCA, which explores the SAR images generated from previous iterations and consequently overcomes, in terms of detection, those noniterative CD methods based on SAR image pairs.
2) Our iterative method demonstrates that, using only two real SAR images, it is possible to obtain results comparable to those of the reference RPCA-based CD method applied to SAR image stacks.
3) The proposed iterative approach is compatible with both the matrix-based RPCA and its tensor extension, TRPC, enabling its application to multitemporal and multidimensional SAR data.
4) The step size of the iterative CD method enables the use of different amounts of reference images, which gradually suppresses false alarms and consequently obtains better performance.

It is worth emphasizing that the proposed method differs from recent artificial-intelligence-based approaches in terms of data requirements and practical applicability. Although AI-based methods have shown promising performance for SAR CD, they generally require large amounts of representative training data, reliable labels, and careful network training. These requirements can be restrictive in practical SAR scenarios, especially when only a small number of well-coregistered acquisitions are available or when labeled change/no-change samples are scarce. On the contrary, the proposed method is fully unsupervised and requires no training samples, making it directly applicable to the available SAR images. Therefore, the proposed iterative RPCA-based scheme is particularly suitable for small SAR datasets, where acquiring large multitemporal stacks or building labeled training sets may be costly or impractical.

The rest of the article is presented as follows. Section II describes the RPCA solved via PCP and the tensor version, TRPCA, solved via the new TNN. The proposed iterative method, based on blind signal decomposition techniques, is presented in Section III, along with the iterative process and implementation aspects of RPCA and TRPCA. Section IV presents the dataset used, the evaluation metrics of the proposed method, and the main results using both techniques. Finally, the final remarks are mentioned in Section V.

II. Robust PCA

PCA is considered one of the most effective unsupervised techniques for feature extraction and dimensionality reduction of large datasets in data analysis and machine learning applications [46]. PCA seeks to orthogonally transform an original dataset, usually highly correlated and linear, into a new dataset with fewer dimensions (e.g., fewer variables), with minimal loss of information. The new variables generated are called PCs [47].

PCA assumes that the PCs are linearly uncorrelated, which does not guarantee data separation [48]. In this context, many RPCA techniques have been developed [40], [41], [48]. RPCA is a blind signal separation technique that typically exploits two types of information in the decomposition: low-rank and sparse components. In this case, obtaining theoretical guarantees that ensure signal separation is even possible. Furthermore, the consideration of sparse terms makes RPCA less sensitive to missing data and outliers compared to PCA [49].

In this scenario, Candés et al. [49] showed that both low-rank and sparse components could be recovered even if the rank grows almost linearly in the dimension of the data and the errors in the sparse content, such as outliers and missing data, are up to a constant fraction of all entries. This problem is known as PCP, and the following convex optimization gives it:

$$\begin{aligned} &\text{minimize} \quad \|\boldsymbol{L}\|_* + \lambda_{\text{RPCA}} \|\boldsymbol{S}\|_1 \\ &\text{s.t.} \quad \boldsymbol{L} + \boldsymbol{S} = \boldsymbol{X} \end{aligned} \tag{1}$$

where $\boldsymbol{X}$, $\boldsymbol{L}$, and $\boldsymbol{S} \in \mathbb{R}^{n_1 \times n_2}$ are the original, low rank, and sparse matrix, respectively. In this formulation, the sum of the singular values σ_k (nuclear norm) is considered as a low-rank approximation, i.e., $\|\boldsymbol{L}\|_* = \sum_k \sigma_k(\boldsymbol{X})$ whereas sparse component is recovered based on the sum of absolute values of the entries (norm l_1), i.e., $\|\boldsymbol{S}\|_1 = \sum_{ij} |\boldsymbol{X}_{ij}|$, with i and j representing the all entries [49]. The content in both matrices is balanced by the parameter λ_{RPCA} given by

$$\lambda_{\text{RPCA}} = \frac{1}{\sqrt{\max(n_1, n_2)}} \tag{2}$$

where n_1 and n_2 are the rows and columns of $\mathbf{X}$, respectively. According to [49], this value is a suggestion and

can be adjusted depending on the application. For example, Jiang et al. [50] showed that $1.5\lambda_{\text{RPCA}}$ is the best parameter for recovering color images. Schwartz et al. [31] shows that $9\lambda_{\text{RPCA}}$ increases the probability of detection of camouflaged vehicles in SAR images, or even Moreira et al. [51] shows that a better classification accuracy on oil platforms is obtained from a $0.1\lambda_{\text{RPCA}}$.

Using a matrix approach may not be an effective strategy in some applications, such as those using hyperspectral images or CD and GMTI in SAR images. The matrix approach requires a vectorization of images, which breaks the intrinsic structure of the data, reducing the effectiveness of the algorithm [52]. To deal with the multidimensional structure of data, many studies have addressed the use of tensors instead of matrices [34], [53], [54], [55], [56].

Due to the lack of a unified definition of tensor rank, different tensor ranks and their relaxations have been proposed [57]. Recently, Lu et al. [39] proposed a solution to TRPCA. In this work, they rigorously defined the tensor spectral norm, tensor average rank, and the new TNN. These norms are based on the definition of the tensor–tensor product (t-product) under DFT, T-SVD [58], which is a new tensor decomposition method similar to the SVD in matrices and its induced tensor tubal rank [59]. With a recovery guarantee under certain conditions [39], TRPCA via TNN decomposes a tensor data $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{m_1 \times m_2 \times m_3}$ into low-rank $\boldsymbol{\mathcal{L}}$ and sparse tensor $\boldsymbol{\mathcal{S}}$ both $\in \mathbb{R}^{m_1 \times m_2 \times m_3}$, through the following optimization problem:

$$\begin{aligned} &\text{minimize} \quad \|\boldsymbol{\mathcal{L}}\|_* + \lambda_{\text{TRPCA}} \|\boldsymbol{\mathcal{S}}\|_1 \\ &\text{s.t.} \quad \boldsymbol{\mathcal{L}} + \boldsymbol{\mathcal{S}} = \boldsymbol{\mathcal{X}} \end{aligned} \tag{3}$$

where $\|\boldsymbol{\mathcal{L}}\|_*$ is the new tensor nuclear norm [60], computed by the sum of singular values along the third dimension of $\boldsymbol{\mathcal{L}}$, i.e., $\|\boldsymbol{\mathcal{L}}\|_* := \sum_{i=1}^{m_3} \|\overline{\boldsymbol{\mathcal{L}}}^{(i)}\|_*$, with $\overline{\boldsymbol{\mathcal{L}}}$ being the result of fast Fourier transformation (fft) of $\boldsymbol{\mathcal{L}}$ along the third dimension, i.e., $\overline{\boldsymbol{\mathcal{L}}} = \text{fft}(\boldsymbol{\mathcal{L}}, [], 3)$ [39], [59], $\|\boldsymbol{\mathcal{S}}\|_1$ is the norm l_1 used in the RPCA case but applied to tensors instead, and λ_{TRPCA} is a weighted parameter balancing both components given by

$$\lambda_{\text{TRPCA}} = \frac{1}{\sqrt{\max(m_1, m_2)\, m_3}} \tag{4}$$

where m_1, m_2, and m_3 are the horizontal, lateral, and frontal slices of $\boldsymbol{\mathcal{X}}$, respectively. Similar to the RPCA [49], TRPCA can be solved by convex optimization techniques because the TNN is the tightest convex relaxation for the tensor averaged rank within the unit ball of the tensor spectral norm [39].

Compared with standard RPCA, TRPCA provides a more natural representation for multidimensional SAR data. In matrix-based RPCA, a stack of SAR images is typically rearranged into a matrix ($\boldsymbol{X} \in \mathbb{R}^{N \times nm}$), where each SAR image is vectorized into a 1-D observation. Although this representation enables matrix-based low-rank and sparse decomposition, it may partially discard the data's original multidimensional structure. On the contrary, TRPCA preserves the SAR data as a tensor ($\boldsymbol{\mathcal{X}} \in \mathbb{R}^{N \times n \times m}$), maintaining the relationship between the image index and the spatial dimensions. This tensor representation allows the low-rank background and sparse change components to be modeled while preserving multiway correlations in the data. Therefore, TRPCA is particularly advantageous for multidimensional SAR datasets, such as multitemporal, multichannel, polarimetric, or volumetric SAR stacks.

Algorithm 1 Solve (1) by ADMM [49], [62]

Input: $\boldsymbol{X}$, $\boldsymbol{L}^{(0)} = \boldsymbol{S}^{(0)} = \Lambda^{(0)} = 0$, $\epsilon > 0, \rho > 1$, $\mu_{max} \geq \mu^0 > 0$.
Output: $\boldsymbol{L}, \boldsymbol{S}$.
repeat
$\boldsymbol{L}^{(t+1)} = \mathcal{D}_{\frac{1}{\mu^{(t)}}}\left(\boldsymbol{X} - \boldsymbol{S}^{(t)} + \frac{\Lambda^{(t)}}{\mu^{(t)}}\right)$;
$\boldsymbol{S}^{(t+1)} = \mathcal{E}_{\frac{\lambda_{\text{RPCA}}}{\mu^{(t)}}}\left(\boldsymbol{X} - \boldsymbol{L}^{(t+1)} + \frac{\Lambda^{(t)}}{\mu^{(t)}}\right)$;
$\Lambda^{(t+1)} = \Lambda^{(t)} + \mu^{(t)}\left(\boldsymbol{S}^{(t+1)} + \boldsymbol{L}^{(t+1)} - \boldsymbol{X}\right)$.
$\mu^{(t+1)} = \min(\rho\mu^{(t)}, \mu_{max})$.
until $\frac{\|\boldsymbol{X} - \boldsymbol{L}^t - \boldsymbol{S}^t\|_F}{\|\boldsymbol{X}\|_F} \leq \epsilon$.

Algorithm 2 Solve (3) by ADMM [39], [60], [67]

Input: $\boldsymbol{\mathcal{X}}$, $\boldsymbol{\mathcal{L}}^{(0)} = \boldsymbol{\mathcal{S}}^{(0)} = \boldsymbol{\mathcal{Y}}^{(0)} = 0$, $\epsilon > 0, \rho > 1$, $\mu_{max} \geq \mu^0 > 0$.
Output: $\boldsymbol{\mathcal{L}}, \boldsymbol{\mathcal{S}}$
repeat
$\boldsymbol{\mathcal{L}}^{(t+1)} = \mathcal{T}_{\frac{1}{\mu^{(t)}}}\left(\boldsymbol{\mathcal{X}} - \boldsymbol{\mathcal{S}}^{(t)} + \frac{\boldsymbol{\mathcal{Y}}^{(t)}}{\mu^{(t)}}\right)$;
$\boldsymbol{\mathcal{S}}^{(t+1)} = \mathcal{E}_{\frac{\lambda_{\text{TRPCA}}}{\mu^{(t)}}}\left(\boldsymbol{\mathcal{X}} - \boldsymbol{\mathcal{L}}^{(t+1)} + \frac{\boldsymbol{\mathcal{Y}}^{(t)}}{\mu^{(t)}}\right)$;
$\boldsymbol{\mathcal{Y}}^{(t+1)} = \boldsymbol{\mathcal{Y}}^{(t)} + \mu^{(t)}\left(\boldsymbol{\mathcal{S}}^{(t+1)} + \boldsymbol{\mathcal{L}}^{(t+1)} - \boldsymbol{\mathcal{X}}\right)$;
$\mu^{(t+1)} = \min(\rho\mu^{(t)}, \mu_{max})$.
until $\|\boldsymbol{\mathcal{L}}^{t+1} - \boldsymbol{\mathcal{L}}^t\|_\infty \leq \epsilon$ **or** $\|\boldsymbol{\mathcal{S}}^{t+1} - \boldsymbol{\mathcal{S}}^t\|_\infty \leq \epsilon$ **or** $\|\boldsymbol{\mathcal{L}}^{t+1} + \boldsymbol{\mathcal{S}}^{t+1} - \boldsymbol{\mathcal{X}}\|_\infty \leq \epsilon$.

There are different methods for solving convex optimization problems, such as those found in the RPCA (1) and TRPCA problems (3), with a theoretical convergence guarantee [61]. Since there is an additive mixture and equality constraints, the alternating direction multipliers method [62] becomes very attractive [63], [64], [65]. The alternating direction method of multiplier (ADMM) is a generalization of the augmented Lagrange method (ALM). However, the ADMM exploits the additive structure and allows the variables to be updated separately, thereby dealing with smaller, simpler subproblems [62].

The detailed algorithm of the ADMM method applied to RPCA [49] and TRPCA [39] can be seen in Algorithm 1 and Algorithm 2, respectively. In these algorithms, Λ and $\boldsymbol{\mathcal{Y}}$ are denoting the Lagrange multiplier for RPCA and TRPCA, respectively, $\mu > 0$ denotes the incoherence parameter [62], ρ can be considered a step size to increase μ to μ_{max} and, consequently, improve the convergence rate, $\mathcal{E}_k$ is the soft-thresholding operator, i.e., proximal operator of norm l_1) [49], $\mathcal{D}_k$ is the singular value thresholding operator [66], i.e., $\mathcal{E}_k$ applied to the singular values of the SVD, and $\mathcal{T}_k$ is the tensor singular value thresholding operator detailed in [39].

III. Iterative Proposed Method

The proposed iterative CD method is presented in Fig. 1. The method aims to provide more information to the robust tool by increasing the number of reference images in the stack as λ increases. Thus, targets in the surveillance image will become sparse in the stack, resulting in better detection.

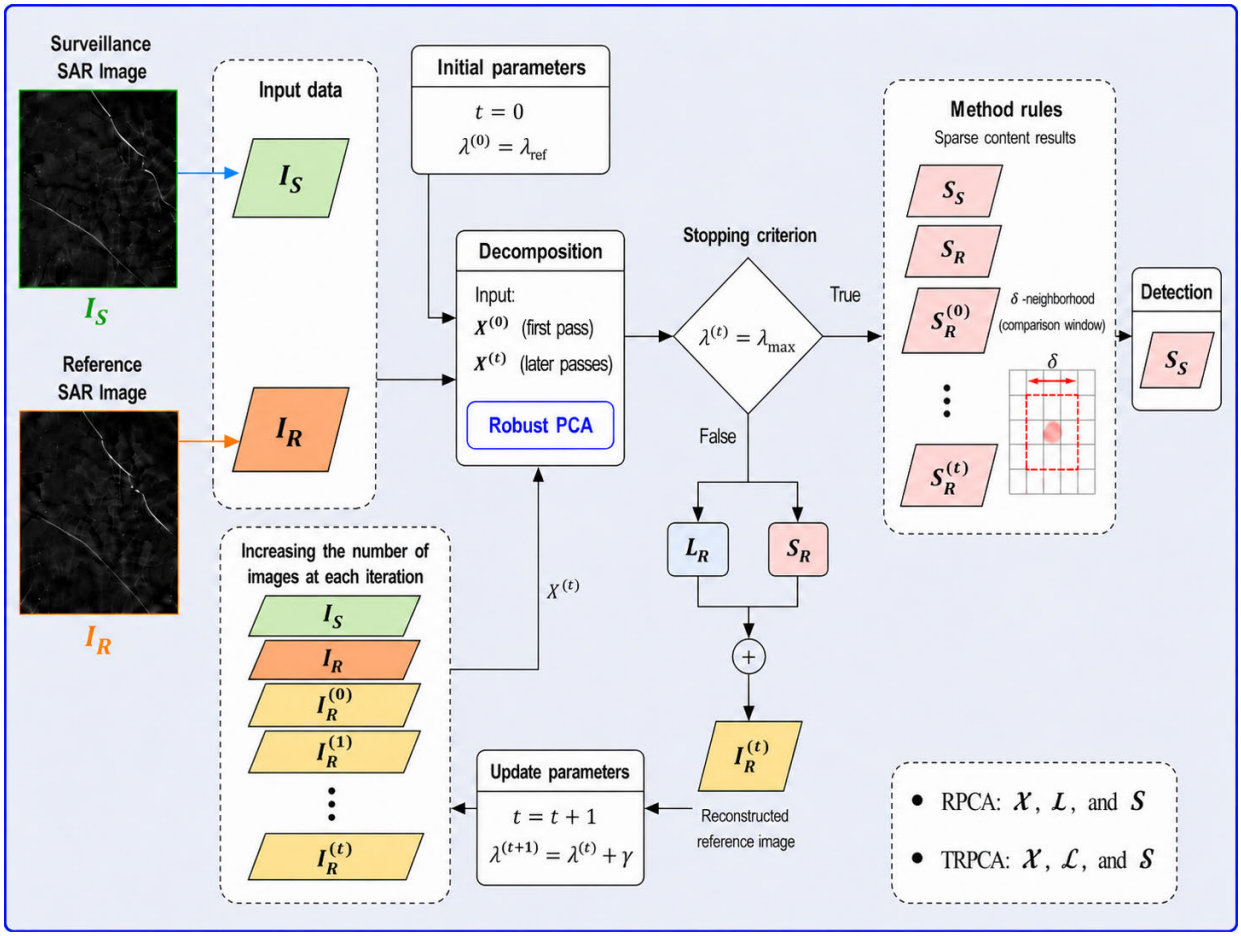


Fig. 1. Iterative CD method based on robust PCA approaches.

Let $I_S \in \mathbb{R}^{n\times m}$ be the surveillance image the one to detect targets and $I_R \in \mathbb{R}^{n\times m}$ be the reference image, where n and m columns denote the rows and columns of the image, respectively. The iterative method can be considered in different approaches (matrix or tensor); the difference lies in how the data is organized.

For the matrix approach (i.e., RPCA case), we consider $\boldsymbol{X} \in \mathbb{R}^{N\times nm}$ formed by the concatenation of N SAR images. This step involves concatenating the rows of the SAR image. Therefore, $n_1 = N$, $n_2 = nm$, and $nm \gg N$, λ_{RPCA} (2) can be approximated by

$$\lambda_{RPCA} = \frac{1}{\sqrt{\max(N, nm)}} = \frac{1}{\sqrt{nm}}. \tag{5}$$

For the tensor approach (i.e., TRPCA case), there are different combinations to form the tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{m_1\times m_2\times m_3}$. They differ in the way t-SVD is obtained [34], [60]. In this iterative CD method, the SAR image stack is represented as a third-order tensor $\boldsymbol{\mathcal{X}} \in \mathbb{R}^{N\times n\times m}$. From this arrangement and considering $n \gg N$, λ_{TRPCA} (4) is given by

$$\lambda_{TRPCA} = \frac{1}{\sqrt{\max(N, n)\, m}} = \frac{1}{\sqrt{nm}}. \tag{6}$$

We evaluated different arrangements for both matrices and tensors and observed no significant performance gains. Therefore, we adopted this strategy because allows a generalized single parameter for both cases since λ_{RPCA} (5) = λ_{TRPCA} (6) which depends only on the size of the SAR image, and it is also considered in other works [31], [32], [67].

Fig. 2 shows how the data is organized in our method for both cases. $\boldsymbol{X}$ is formed by the concatenation the rows of I_S and I_R, i.e., X_S and X_R and $\boldsymbol{\mathcal{X}}$ is formed by the tensorization of I_S and I_R, i.e., $\mathcal{X}_S$ and $\mathcal{X}_R$. The input to the method is just two SAR images, as in traditional CD methods, i.e., $N = 2$. However, N increases over iterations, getting images from previous iterations. This process will be further detailed below.

The core idea of the proposed method is to transform an SAR image-pair CD problem into a progressively enriched reference-data problem. Although the method starts from only one surveillance SAR image (I_S) and one reference SAR image (I_R), additional reference images are generated during the iterative process and included in the input data for the subsequent iterations. More specifically, RPCA/TRPCA is applied to decompose the input data into low-rank and sparse components, and the components associated with the reference image are used to generate a new reference image $I_R^{(t)}$. By progressively increasing the amount of reference information, the method improves the separation between background/reference-related responses and target-related sparse changes. This strategy is particularly useful for small SAR datasets, where large multitemporal stacks are unavailable.

The main difference between traditional RPCA-based CD methods and the proposed scheme is that traditional

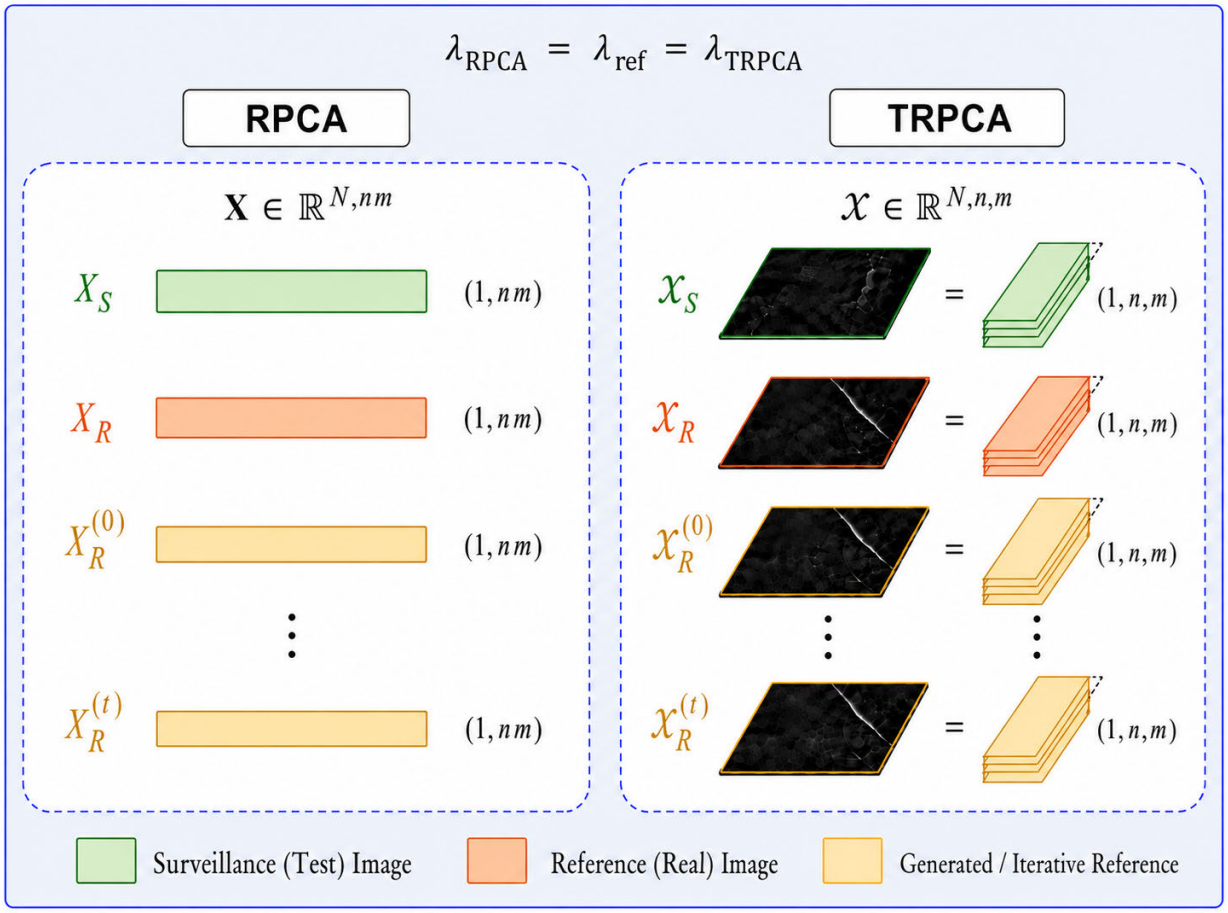


Fig. 2. Representation of SAR images for RPCA and TRPCA implementation.

TABLE I
IMPLEMENTATION PARAMETERS FOR RPCA AND TRPCA

Parameter	Values
λ_{ref}	λ_{RPCA} (2) or λ_{TRPCA}(4)
$\lambda^{(0)}$	λ_{ref}
λ_{max}	[1, 20] λ_{ref}
γ	[0.25, 0.5, 1]
$\mu^{(0)}$	$10\lambda_{ref}$
μ_{max}	10^{10}
ϵ	10^{-6}
ρ	1.1

approaches usually perform a single decomposition of the available SAR images into low-rank and sparse components. The change map is then obtained directly from the sparse component of this single decomposition. On the contrary, the proposed method performs this process iteratively. At each iteration, a new reference image is generated and added to the input data, increasing the amount of reference information available for the next decomposition. This iterative strategy helps reduce reference-related sparse responses and false alarms while preserving target-related changes in the surveillance image.

A. Iterative Process

Initially, select the surveillance and reference SAR image to create $\mathrm{X}^{(t)}$ (i.e., either $\boldsymbol{X}$ or $\boldsymbol{\mathcal{X}}$), the λ value of the first iteration, i.e., $\lambda^{(0)}$. In this study, we consider $\lambda^{(0)} = \lambda_{ref}$ defined by (5) and (6). Furthermore, define a maximum value for λ, i.e., λ_{max}. This value serves as the stopping criterion for the iterative CD method.

TABLE II
NOTATION OF THE DATASET

Mission	Pass					
	1	2	3	4	5	6
2	M2P1	M2P2	M2P3	M2P4	M2P5	M2P6
3	M3P1	M3P2	M3P3	M3P4	M3P5	M3P6
4	M4P1	M4P2	M4P3	M4P4	M4P5	M4P6
5	M5P1	M5P2	M5P3	M5P4	M5P5	M5P6

The iterative approach consists of applying robust PCA to $\mathrm{X}^{(t)}$. If $\lambda^{(0)} = \lambda_{max}$, the method converges at the first iteration and the method rules (described in Section III-B) are applied in $\mathrm{S}^{(0)}$ which, in this case, is formed by the sparse content from the surveillance $\mathrm{S_S}$ and reference $\mathrm{S_R}$ image.

If $\lambda^{(0)} < \lambda_{max}$, the low-rank ($\mathrm{L_R}$) and sparse ($\mathrm{S_R}$) components of the reference image are added to create an approximation of $\mathrm{I_R}$, i.e., $\mathrm{I_R^{(t)}}$. The difference between the approximation $\mathrm{I_R^{(t)}}$ and the real image $\mathrm{I_R}$ is defined mainly by the error value adopted in the RPCA implementation. Then, we create a new stack $\mathrm{X}^{(t)}$ considering the $\mathrm{I_R^{(t)}}$ generated at the previous iteration and update the parameters t and $\lambda^{(t)}$ for the next iteration. Note that $\mathrm{I_R^{(t)}}$ is always generated from $\mathrm{L_R} + \mathrm{S_R}$, i.e., from the decomposition of reference image $\mathrm{I_R}$.

The method converges when $\lambda^{(t)} = \lambda_{max}$, then the output of the proposed iterative method is $S^{(t)}$ which is formed by $[S_S, S_R, S_R^{(0)}, \ldots, S_R^{(t)}]$, where $S_R^{(0)}, S_R^{(1)}, \ldots, S_R^{(t)}$ is the sparse content regarding $I_R^{(0)}, I_R^{(1)}, \ldots, I_R^{(t)}$, respectively. In the method, $\lambda^{(t)}$ increases from λ_{ref} to λ_{max} with a step size γ, which controls the number of images in the SAR image stack. Lower γ values imply a more extended method convergence time; however, more reference images are in the stack for the same value of $\lambda^{(t)}$ compared to higher γ.

The method exploits the approximation of the reference image from the previous iteration to increase the number of SAR images in the stack, making the targets in the surveillance images sparse in the whole new stack. Thus, we are creating a stack of reference SAR images that changes very little over time, similar to video surveillance applications with optical images, where robust PCA stands out [41].

B. Method Rules

For a fair comparison with other reference SAR CD methods that employ either RPCA or TRPCA, three specific rules, known as "method rules," were applied. The method rules mainly aim to detect targets that appear in the scene (i.e., something that was not observed in the reference content and appears in the surveillance image) and eliminate false alarms that are not related to the spatial resolution of the SAR image and the application of interest, which is target detection, in this case, military vehicles hidden in forested areas. Such rules were proposed for CD methods in SAR images based on RPCA [32] and TRPCA [38] and will be described below.

1) *First Rule:* Only detections in S_S will be counted, as the goal of the CD method is to identify targets present in the surveillance image I_S.
2) *Second Rule:* Only positive values in S_S are treated as detections. Positive results for sparse content correspond to variations that appear in the surveillance image but are absent from the reference images. Negative values in the sparse content typically represent static objects (e.g., buildings, fences) captured in the references but missing in the surveillance image. Since the aim is not to detect objects presented in the reference content, such negative values can be disregarded.
3) *Third Rule:* Discard any detection in S_S that also appears in at least one reference SAR image ($S_R^{(0)}, S_R^{(1)}, \ldots, S_R^{(t)}$) within a $2\delta \times 2\delta$ pixel window. This ensures that detections linked to the same object present in both the surveillance and reference images are removed. As adopted in the reference methods, $\delta = 9$ was used. The choice of this value is related to the spatial resolution of the SAR data from the CARABAS-II system.

C. Implementation Aspects

In this study, the implementations of RPCA and TRPCA presented in [68] and [69], respectively, were considered as references. Therefore, the following values of μ_0, μ_{max}, ϵ, and ρ presented in Table I were used. We chose λ_{ref} based on the RPCA (2) and TRPCA (4) formulations and ranged it up to $\lambda_{max} = 20 \times \lambda_{ref}$. Higher maximum lambda values were tested, and no gains in detection were observed.

We adopted different values for γ, i.e., $\gamma = \{0.25, 0.5, 1\}$. Smaller values were also tested; however, using values below 0.25 increased the number of images in the stack, significantly extending runtime and exceeding the machine's capabilities for handling such large datasets. No significant performance improvement was observed for γ values smaller than 0.25.

IV. Experimental Results

A. CARABAS Data

In this study, the proposed SAR CD method was evaluated using the CARABAS-II SAR dataset [70]. These images were acquired using the Swedish CARABAS-II airborne system, a UWB system operating in the VHF band, offering a resolution of approximately 2.5 meters in both azimuth and range. The images obtained by this type of system are referred to as VHF wavelength-resolution SAR images [71], [72]. These images are characterized by their high temporal stability and by their insensitivity to speckle noise [73]. Their main backscatter originates from large objects, e.g., large trunks and constructed structures, resulting in low sensitivity to small scatterers. This makes them suitable for FOPEN applications [74].

The dataset images were collected during a 2002 flight campaign at RFN Vidsel military base. Each image displays 25 concealed forest canopy targets: 11 small, eight medium, and seven large vehicles. The region includes forests, roads, water bodies, and fields. The dataset includes 24 calibrated, preprocessed, and geocoded wavelength-resolution amplitude images. Each image covers the same 6 km^2 (3 km $\times$2 km) with 1 m^2 pixel resolution. Passes are grouped by flight geometry and the intensity of radio frequency interference. More information on the dataset is available in [74].

In this study, the SAR images are represented using the notation MxPy, where Mx means the mission $x \in \{2, 3, 4, 5\}$ (i.e., deployment of the targets), and Py represents the pass $y \in \{1, 2, 3, 4, 5, 6\}$. Table II presents the notation for each image of the CARABAS data. Examples of SAR images from each mission, along with a map of the imaged region, are presented in Fig. 3.

B. Evaluation Metrics

Two metrics were used to evaluate the proposed iterative CD method: PD and FAR. Those metrics are commonly used to assess CD methods, such as the one that uses the CARABAS dataset [15], [17], [18], [22], [32], [38], and are defined by

$$\text{PD} = \frac{\text{DT}}{\text{KT}}, \quad \text{FAR} = \frac{\text{FA}}{\text{IA}} \tag{7}$$

where DT denotes the number of targets detected in the surveillance image, KT represents the number of known targets in the scene (for the CARABAS-II dataset, each surveillance image contains 25 hidden military vehicles, so KT = 25), FA is the number of false alarms in the surveillance image (i.e., number of detections not related to the targets ground truth), and IA corresponds to the area covered by the SAR image, which is 6 km^2.

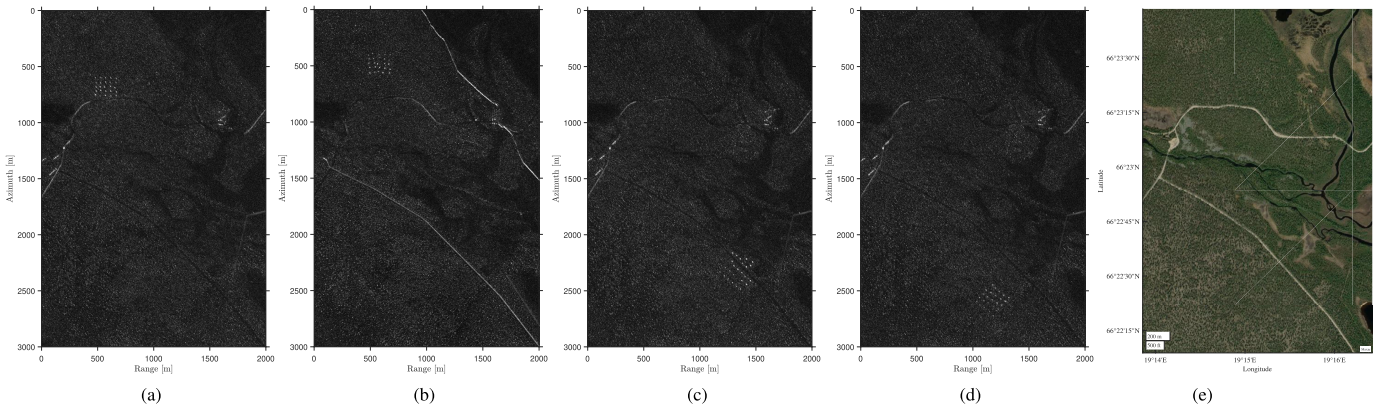


Fig. 3. Example of magnitude HH-polarized SAR images from the CARABAS-II dataset and a map covering the same area as the images. (a) M2P1 (225°). (b) M3P2 (135°). (c) M4P5 (230°). (d) M5P5 (230°). (e) Map of the region, 2024.

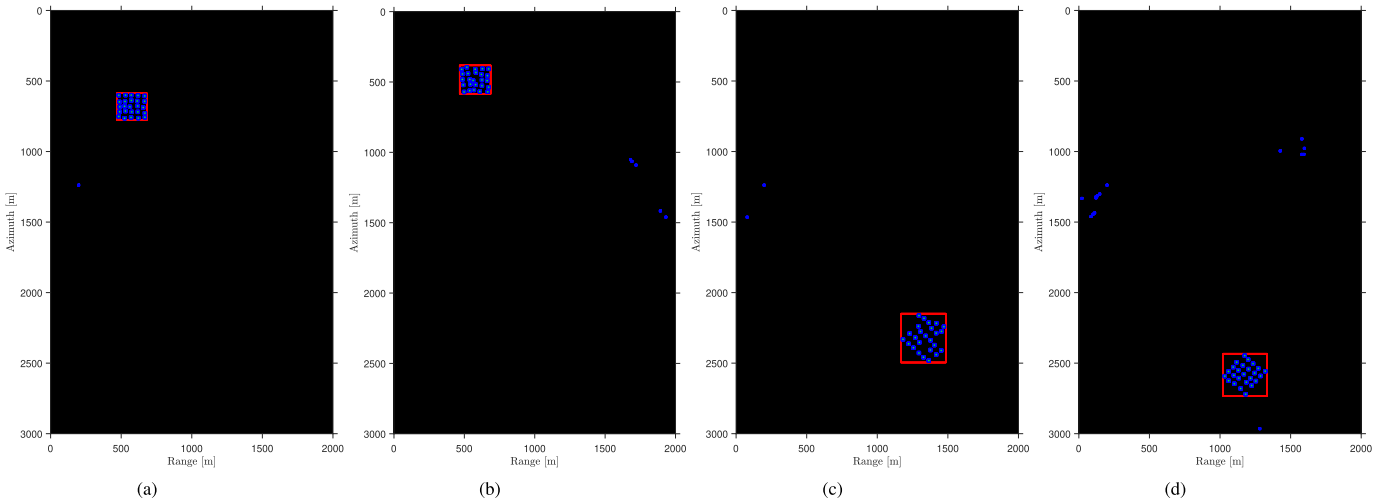


Fig. 4. Iterative CD-RPCA results for $\lambda_{max} = 8\lambda_{ref}$ and $\gamma = 0.25$. $\mathrm{I_S}$ is the surveillance SAR image, and $\mathrm{I_R}$ is the reference image. Blue dots inside the red square are the ground truth, with green dots being the detected targets in the surveillance image presented in Fig. 3. Blue dots outside the red square are false alarms. (a) $\mathrm{I_S}$:M2P1 and $\mathrm{I_R}$:M3P1. (b) $\mathrm{I_S}$:M3P2 and $\mathrm{I_R}$:M5P2. (c) $\mathrm{I_S}$:M4P5 and $\mathrm{I_R}$:M2P5. (d) $\mathrm{I_S}$:M5P5 and $\mathrm{I_R}$:M3P5.

C. CD Results

Initially, the iterative CD method was evaluated by using the RPCA formulation. Since we are focused on proposing a new CD method, we aim to validate it using RPCA and then demonstrate its extension to tensors.

Fig. 4 shows the detection results considering surveillance images from different missions, $\lambda_{max} = 8\lambda_{ref}$, and $\gamma = 0.25$. In Fig. 4(a), we can observe the 25 detected targets and one false alarm present in the scene. On the other hand, Fig. 4(d) also shows the 25 detected targets, but with 14 false alarms. Those false alarms are related to structures along the sensitive flight path, such as fences and power lines, which are shown in Fig. 3(d), azimuth 1000–1700.

Table III provides an extension of the visual results shown in Fig. 4, presenting the PD and FAR metrics for all SAR image pairs considered in the study and different γ values. Those pairs are the same ones adopted by the CARABAS-II dataset challenge problem. In this scenario, the best result is obtained using $\gamma = 0.25$, with a PD of approximately 97% and 74 FA, i.e., FAR = 0.514 FA/km^2. Note that by changing γ to 1, the number of FA increased by 24.3%, from 74 to 92, maintaining the same PD. The results show that our iterative CD method, based on pairs of SAR images as input, already achieves superior performance compared to CD methods for image pairs, such as the CD challenge problem [74], where a PD of 97% and FAR = 0.650 FA/km^2 are obtained.

Overall, the step size γ controls the tradeoff between false alarm reduction and computational cost. Smaller γ values lead to more iterations and produce more reference images, providing more reference information for the RPCA/TRPCA decomposition. This tends to reduce the FAR while preserving a high detection probability. On the other hand, larger γ values reduce the number of iterations and computational cost but may provide less reference information, thereby resulting in a higher FAR.

Note that a γ reduction results in a greater M, i.e., there are more reference images, as previously stated. This increase in reference content means that more information is provided to the RPCA; as a result, the surveillance image targets become sparser. In addition, the FA content from fences and transmission lines, which are generally stationary over time, can be better represented in the low-rank component of the

TABLE III
ITERATIVE CD-RPCA RESULTS FOR $\lambda_{max} = 8\lambda_{ref}$

SAR Images			$\gamma = 0.25$, $t = 27$			$\gamma = 0.50$, $t = 13$			$\gamma = 1$, $t = 6$			CD Challenge Problem [74]		
Surveillance I_S	Reference I_R	KT	DT	PD	FA	DT	PD	FA	DT	PD	FA	DT	PD	FA
M2P1	M3P1	25	25	1.000	1	25	1.000	1	25	1.000	1	25	1.000	2
M3P1	M4P1	25	23	0.920	3	23	0.920	3	23	0.920	3	22	0.880	1
M4P1	M5P1	25	25	1.000	0	25	1.000	0	25	1.000	0	25	1.000	2
M5P1	M2P1	25	25	1.000	6	25	1.000	6	25	1.000	6	23	0.920	4
M2P2	M4P2	25	25	1.000	2	25	1.000	2	25	1.000	2	25	1.000	2
M3P2	M5P2	25	25	1.000	1	25	1.000	3	25	1.000	5	25	1.000	4
M4P2	M2P2	25	25	1.000	4	25	1.000	5	25	1.000	8	25	1.000	3
M5P2	M3P2	25	25	1.000	4	25	1.000	5	25	1.000	7	25	1.000	4
M2P3	M5P3	25	25	1.000	3	25	1.000	3	25	1.000	3	25	1.000	3
M3P3	M2P3	25	23	0.920	1	23	0.920	1	23	0.920	1	23	0.960	4
M4P3	M3P3	25	25	1.000	2	25	1.000	2	25	1.000	2	25	1.000	0
M5P3	M4P3	25	25	1.000	0	25	1.000	0	25	1.000	0	24	0.960	2
M2P4	M3P4	25	25	1.000	3	25	1.000	3	25	1.000	2	24	0.960	3
M3P4	M4P4	25	25	1.000	2	25	1.000	3	25	1.000	2	25	1.000	2
M4P4	M5P4	25	25	1.000	5	25	1.000	5	25	1.000	6	25	1.000	4
M5P4	M2P4	25	24	0.960	8	24	0.960	9	24	0.960	12	25	1.000	4
M2P5	M4P5	25	25	1.000	0	25	1.000	0	25	1.000	1	25	1.000	3
M3P5	M5P5	25	15	0.600	12	15	0.600	12	15	0.600	12	17	0.680	10
M4P5	M2P5	25	25	1.000	1	25	1.000	2	25	1.000	2	25	1.000	2
M5P5	M3P5	25	25	1.000	14	25	1.000	14	25	1.000	15	23	0.920	29
M2P6	M5P6	25	25	1.000	2	25	1.000	6	25	1.000	2	25	1.000	1
M3P6	M2P6	25	24	0.960	0	25	1.000	0	24	0.960	0	25	1.000	3
M4P6	M3P6	25	25	1.000	0	25	1.000	0	25	1.000	0	25	1.000	1
M5P6	M4P6	25	25	1.000	0	25	1.000	0	25	1.000	0	23	0.920	3
Total		**600**	**584**	**0.973**	**74**	**585**	**0.975**	**85**	**584**	**0.973**	**92**	**579**	**0.965**	**96**

TABLE IV
BEST RESULTS OF OUR ITERATIVE CD METHOD BASED ON TRPCA AND COMPARISON WITH THE NONITERATIVE CD METHOD BASED ON TRPCA [38]

Method	PD [%]	FAR [FA/km^2]
$N = 2$, CD-TRPCA [38]	82.2	1.240
$N = 2$, Iterative CD-TRPCA, $\gamma = 1$	82.5	0.778
$N = 2$, Iterative CD-TRPCA, $\gamma = 0.5$	82.6	0.687
$N = 2$, Iterative CD-TRPCA, $\gamma = 0.25$	82.6	0.590
$N = 7$, CD-TRPCA [38]	92.9	0.523

RPCA, and thus, a reduction in FA in the sparse matrix is obtained.

The performance gain for small values of γ is also explained by the method's step rules, which discard detections that occur in the surveillance and reference images within a given window. Since more reference images are included in the stack, the sparse component contains more information about the FA, and consequently, the FA of the surveillance image is reduced.

To better observe the performance of the iterative CD method based on RPCA, we present ROC curves in Fig. 5 for different γ values, varying λ_{max}. For comparison, we present the performance of the reference RPCA-based CD method for pairs and stacks of SAR images presented in [31] and [32]. Note that it was considered the same image pairs used in [31] and [74].

According to Table III and confirmed in Fig. 5, the iterative CD method based on RPCA with $\gamma = 0.25$ showed better performance for different values of λ_{max}. In this context, it was possible to obtain a PD = 95% and FAR = 0.389 FA/km^2 for $\lambda_{max} = 8.75\lambda_{ref}$. For a PD of approximately 90%, this same method achieved a FAR of 0.271 FA/km^2 for $\lambda_{max} = 9.75\lambda_{ref}$.

The iterative CD method outperformed the reference RPCA-based CD method across all scenarios for pairs of SAR images. This shows that an iterative RPCA approach can be an efficient strategy for reducing false alarms in RPCA-based CD methods, thereby improving performance. Our method still performed competitively with the CD-RPCA stack method. In some regions, such as 85% < PD < 94%, it was possible to achieve better FAR performance than with the stack method, even with two input SAR images.

In fact, the reference images generated by RPCA and used to increase the stack in the iterative CD method are very close to the real reference image. In the stack-based RPCA method, the reference images in the stack are real images that are not always similar to the surveillance image. This may better represent the surveillance-image targets in the sparse matrix in the iterative CD method, since the reference images in the iterative CD-RPCA method are more correlated than the real SAR image stack in the reference CD-RPCA method.

This behavior becomes more evident when we analyze the rank of the low-rank component of the RPCA (i.e., $\boldsymbol{L}$) in both methods. In our method, even with many images in the stack and varying values of λ_{max}, the rank of $\boldsymbol{L}$ is always 2, i.e., the generated reference SAR images are linear combinations of the original reference SAR image. In the reference RPCA-based CD method for SAR image stacks, as λ increases, the rank of $\boldsymbol{L}$ remains equal to that of $\boldsymbol{X}$ (input data), i.e., this component exhibits no dimensionality reduction.

There is no significant gain when comparing the results of the iterative CD method based on RPCA (i.e., $N = 2$) and the reference RPCA-based CD method on SAR image

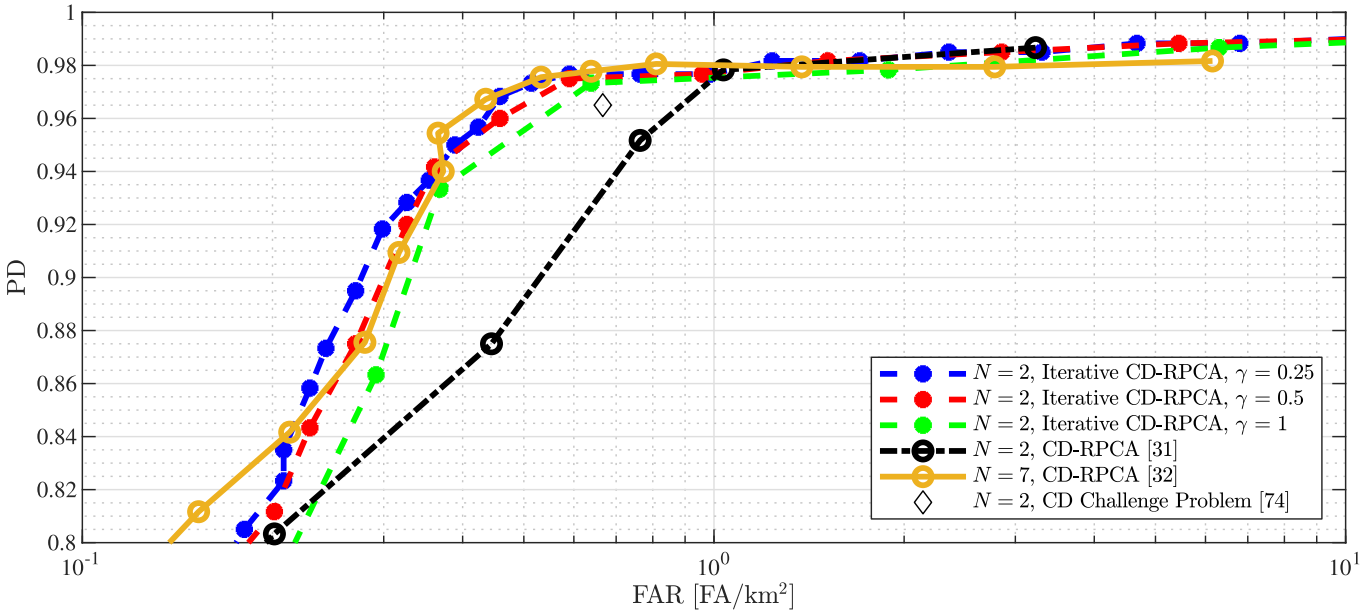


Fig. 5. ROC performance of the iterative CD-RPCA method using different step sizes γ and other reference CD methods.

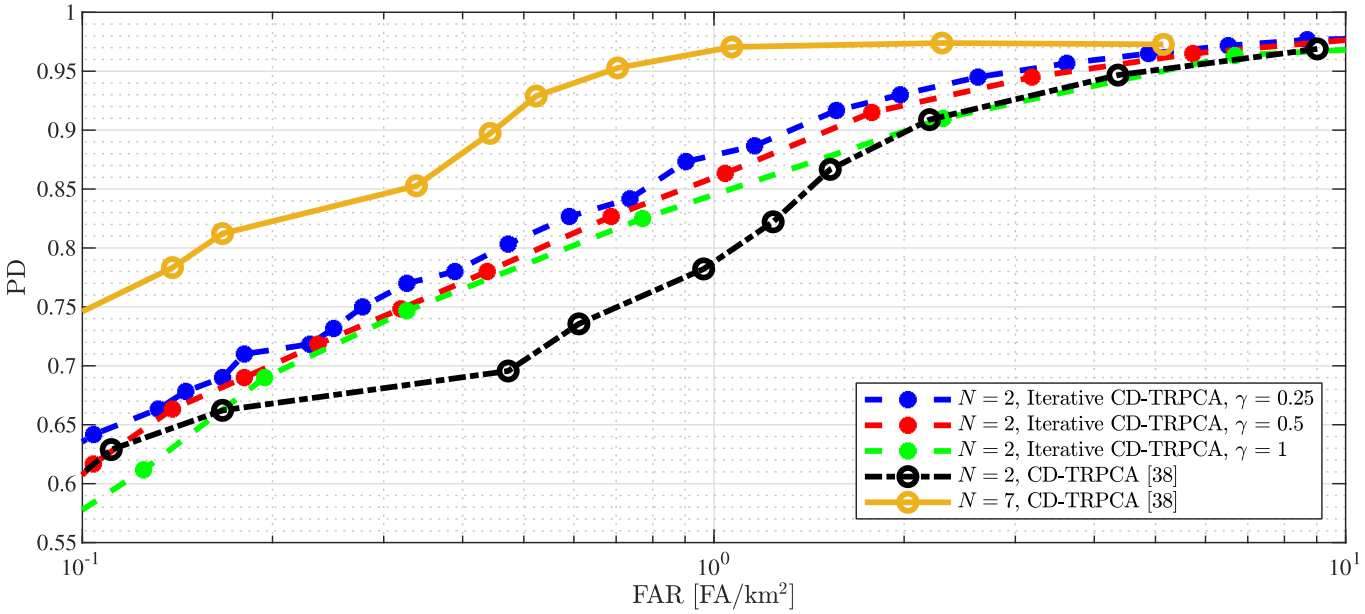


Fig. 6. ROC performance of the iterative CD-TRPCA method using different step sizes γ and the reference CD methods.

stacks (i.e., $N = 7$) [32]. In fact, additional images can help the CD method; however, these gains are small compared to the computational cost. If the goal is computational efficiency, adopting the iterative CD method, which uses less data and processing, is more feasible.

We have extended our iterative CD method to tensors. Fig. 6 shows the results obtained in terms of PD and FAR and compares them with the reference CD method based on TRPCA, which is not iterative [38]. As in the case of RPCA, the iterative CD method performs better in terms of PD and FAR than the noniterative reference CD method based on TRPCA for N = 2. For a PD of about 82%, there was a reduction in FAR from 1.240 FA/km^2 to 0.590 when using the iterative CD method based on TRPCA with $\gamma = 0.25$, as presented in Table IV.

Note that now the iterative method based on TRPCA did not achieve results similar to those obtained with $N = 7$, as was the behavior shown in Fig. 5 for the RPCA case. Indeed, TRPCA explores temporal and spatial correlations across multiple SAR images, which may enable much more detailed and robust modeling of the low-rank component. When generating reference images using the iterative method based on TRPCA, it is very difficult to generate reference images ($\widehat{\mathrm{I}}_{\mathrm{R}}$) similar to the original ones with only two as input.

On the other hand, RPCA works with matrices and relies only on the difference between the images to identify changes, so that the separation into low-rank and sparse components can be sufficiently done with a few images.

This difference is clearly observed in the CD methods presented in [32] and [38]. For a pair of SAR images, the RPCA-based CD method outperforms the TRPCA-based approach. However, as the number of images increases, TRPCA surpasses RPCA in both PD and FAR. For instance, when $N = 19$, the noniterative TRPCA-based CD method [38] achieves significantly better performance than its noniterative RPCA-based counterpart [32].

V. Conclusion

This article proposes an iterative change detection method based on robust PCA for wavelength-resolution SAR image pairs. The proposed method exploits increasing λ up to λ_{max} to increase the number of correlated reference images, thereby creating a reference SAR image stack. Increasing the number of SAR reference images will make the targets in the surveillance images sparser, leading to better detection. We considered different step sizes (γ) for increasing λ, and the best results were obtained when more reference images were generated, i.e., lower γ values. Experiments on the CARABAS-II dataset show that our approach outperforms noniterative, pair-based CD methods. When using the RPCA, the iterative CD method even offers superior performance in some regions compared to the noniterative SAR image stack CD method, reaching a PD = 0.968 and FAR = 0.458 FA/km^2. This strategy is well-suited to real-world applications, where the availability of large SAR image stacks for CD is often limited. The proposed iterative CD approach demonstrates that, even with only image pairs, it can achieve performance comparable to, and in some cases competitive with, noniterative CD methods that rely on multitemporal SAR stacks.

Despite the promising results, some limitations should be highlighted. The experimental validation was conducted using the CARABAS-II dataset, which comprises wavelength-resolution VHF SAR images with high temporal stability. Therefore, the performance of the proposed method may depend on the characteristics of the SAR system, the quality of image coregistration, the similarity between acquisition geometries, and the stability of the imaged scene. In scenarios involving stronger speckle noise, lower temporal coherence, significant geometric distortions, or highly dynamic background changes, additional validation may be required. Moreover, using smaller step-size values increases the number of generated reference images and can improve false-alarm suppression, but at the cost of higher computational demand. These aspects should be considered when extending the proposed iterative RPCA/TRPCA-based approach to other SAR sensors, frequency bands, and operational scenarios.

Code Availability

The source code implementing the proposed iterative RPCA/TRPCA-based CD method will be made publicly available upon publication of this article at: https://github.com/lucaspedrosor/iterative-rpca-sar-change-detection

Appendix: Acronyms and Abbreviations

In Nomenclature, we provide a list of acronyms and abbreviations used in this article.

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